

Cyber Threat Detection System

Academic Research, Theoretical Foundations, Regulatory Standards & Reference Links

REPOSITORY

Vivek-Kamannavar/Cyber-Threat

ARCHITECTURAL PROFILE

Unidirectional Data Diode / Air-Gap

DETECTION CORE

Hybrid ML + Sliding Window Heuristics

DOCUMENT STATUS

Verified Reference Compendium

1. Executive Research Abstract & Problem Statement

Modern mission-critical infrastructures—such as nuclear power plants, electric distribution grids, defense installations, and industrial SCADA networks—rely on **hardware-enforced unidirectional security gateways (Data Diodes)** to isolate high-security enclaves from external untrusted networks. While data diodes guarantee physical immunity against remote reverse-channel intrusions, they invalidate the fundamental assumptions of conventional Network Intrusion Detection Systems (NIDS).

The Data Diode NIDS Dilemma: Traditional network intrusion sensors (such as Snort or Zeek in default configurations) rely on bidirectional TCP handshake tracking, TCP state tables, ACK sequence synchronization, and bi-directional session reconstruction. In a hardware data diode, the return optical fiber strand is physically severed. Transmitters emit light via laser diodes, and receivers consume light via photodiodes without electrical feedback. Consequently, threat detection must operate on *asymmetric, unidirectional, and single-direction packet streams* using temporal sliding windows, mathematical entropy, graph analytics, and unsupervised machine learning.

2. Project Detection Engine & Theoretical Foundations

THREAT MODULE	ALGORITHMIC & MATHEMATICAL BASIS	RESEARCH IMPLEMENTATION IN PROJECT	FOUNDATIONAL REFERENCE
Volumetric DDoS & Floods	Shannon Information Entropy $H(X) = -\sum P(x) \log_2 P(x)$ on source/dest IP distributions.	Evaluated over 60s sliding window in <code>engine/features.py</code> . Flat zero entropy indicates single-source brute flood.	Shannon (1948); Feinstein et al. (2003)
Botnet C2 Beaconing	Inter-Arrival Time (IAT) Coefficient of Variation ($CV = \sigma / \mu < 0.22$) & low timing jitter.	Detects periodic automated beaconing intervals without needing payload inspection or C2 return ACKs.	Bilge et al. (ACSAC 2012); Gu et al. (BotMiner 2008)
DNS Exfiltration & DGA	Character n-gram transition probability, bi-gram rarity, and Shannon domain name entropy.	Flags algorithmic domains (e.g., <code>xk99q21mnv78aaq.xyz</code>) and oversized base64/hex DNS tunneling queries.	Yadav et al. (IMC 2010); Antonakakis et al. (USENIX 2012)
Encrypted Malware (TLS)	JA3 & JA3S MD5 fingerprinting of TLS Client Hello (Cipher Suites, Extensions, Elliptic Curves).	Matches encrypted C2 payloads (Cobalt Strike, Metasploit, AsyncRAT) without decrypting payloads.	Althouse et al. (Salesforce 2017); Anderson & McGrew (ACM 2016)
Reconnaissance Port Scans	Directed Graph Fan-Out ($D_{out} > 15$) and bipartite connection mapping via NetworkX.	Detects horizontal and vertical scanning activity across temporal sliding-window graph topologies.	Akoglu et al. (ACM Comput. Surv. 2015); Staniford et al. (Spice/Snort)
Zero-Day Anomaly Detection	Unsupervised Isolation Forest ($n=100$ trees, $contamination=0.03$) on multi-variate vector space.	Identifies multi-feature structural anomalies deviating from baseline normal flow behavior.	Liu, Ting, & Zhou (IEEE ICDM 2008); Scikit-Learn (2011)

3. Foundational Research Papers & Academic Literature

1. Statistical Metric for DDoS Defense Using Information Entropy

Entropy & Volumetric DoS

Feinstein, L., Schnackenberg, D., Balupari, R., & Kindred, D. (DARPA / IEEE DARPA Information Survivability Conference and Exposition, 2003)

Introduced the seminal application of Shannon entropy to packet header attributes for the instant detection of volumetric DDoS attacks. Serves as the mathematical cornerstone for this project's IP entropy calculation in `engine/features.py`.

<https://ieeexplore.ieee.org/document/1199353>

2. DISCLOSURE: Detecting Botnet Command and Control Servers Through Large-Scale NetFlow Analysis

C2 Timing & Beacons

Bilge, L., Balzarotti, D., Kirda, E., & Kruegel, C. (28th Annual Computer Security Applications Conference - ACSAC 2012)

Formulated the timing regularity algorithm and inter-arrival time coefficient of variation ($CV = \sigma / \mu$) used to differentiate human web browsing from automated botnet C2 heartbeat signals. Implemented in this project's beaconing detection rule.

<https://doi.org/10.1145/2420950.2420969>

3. Detecting Algorithmically Generated Malicious Domain Names

DNS N-Gram & DGA

Yadav, S., Reddy, A. K. K., Reddy, A. L. N., & Ranjan, S. (ACM Internet Measurement Conference - IMC 2010)

Demonstrated that pseudo-random domain generation algorithms (DGAs) can be exposed by measuring character bigram distributions and linguistic entropy against benign Top-1M dictionaries. Directly utilized in this project's DNS n-gram scoring engine.

<https://doi.org/10.1145/1879141.1879148>

4. Isolation Forest: Fast Anomaly Detection for Large-Scale Data

Unsupervised ML

Liu, F. T., Ting, K. M., & Zhou, Z.-H. (Eighth IEEE International Conference on Data Mining - ICDM 2008)

Established the Isolation Forest algorithm, which exploits two quantitative properties of anomalies: few in quantity and attribute-divergent. Forms the core of Tier-2 anomaly detection in `engine/models/isolation_forest.py`.

<https://doi.org/10.1109/ICDM.2008.17>

5. Identifying Encrypted Malware Traffic with Contextual Flow Data

TLS Fingerprinting

Anderson, B., & McGrew, D. (ACM Workshop on Artificial Intelligence and Security - AISEC 2016 / Cisco Research)

Published the methodology of inspecting unencrypted TLS Client Hello parameters (ciphers, extensions, curves) to identify malicious TLS agents without breaking end-to-end encryption. Provides the scientific validation for this repository's JA3 matching.

<https://doi.org/10.1145/2996758.2996768>

6. Graph-Based Anomaly Detection and Graph Mining: A Survey

Graph NetworkX

Akoglu, L., Tong, H., & Koutra, D. (ACM Computing Surveys, Vol. 47, No. 3, 2015)

Synthesized graph-theoretic approaches for modeling network interactions as bipartite graphs. Directly applied in this system using NetworkX to track IP-to-port fan-out ratios for detecting vertical and horizontal reconnaissance scans.

<https://doi.org/10.1145/2701127>

4. Industrial Standards & Regulatory Frameworks for Data Diodes

NIST Special Publication 800-82, Revision 2: Guide to Industrial Control Systems (ICS) Security

NIST SP 800-82

National Institute of Standards and Technology (U.S. Department of Commerce)

Defines standard cybersecurity architectures for SCADA, DCS, and PLC systems. Explicitly recommends unidirectional security gateways (Section 5.4) to segment safety-critical control networks from enterprise systems.

<https://csrc.nist.gov/publications/detail/sp/800-82/rev-2/final>

IEC 62443: Security for Industrial Automation and Control Systems (IACS)

IEC 62443

International Electrotechnical Commission (IEC) / ISA-99 Standards Committee

Defines Zone and Conduit security boundaries (IEC 62443-3-2 and 62443-3-3). Specifies unidirectional conduits as the primary control mechanism for Security Level 4 (SL-4) air-gapped protection against state-sponsored intrusions.

<https://www.isa.org/standards-and-publications/isa-standards/isa-iec-62443-series>

NERC CIP-005-5: Cyber Security — Electronic Security Perimeter(s)

NERC CIP

North American Electric Reliability Corporation (Mandatory Critical Infrastructure Protection)

Mandates strict access control and perimeter isolation for Bulk Electric Systems (BES). Cites unidirectional gateways as a primary compliant architecture for transmitting telemetry out of transmission control centers without inbound ingress.

<https://www.nerc.com/pa/Stand/Reliability%20Standards/CIP-005-5.pdf>

5. Benchmark Datasets, Open Protocols & Core Tooling

DATASET / STANDARD	ORGANIZATION & CITATION	OFFICIAL REFERENCE / SOURCE LINK
CIC-IDS2017 Dataset	Canadian Institute for Cybersecurity (UNB) <i>Sharafaldin et al. (ICISSP 2018)</i>	https://www.unb.ca/cic/datasets/ids-2017.html
UNSW-NB15 Dataset	Australian Defence Force Academy (ADFA) <i>Moustafa & Slay (MilCIS 2015)</i>	https://research.unsw.edu.au/projects/unswnb15-dataset
Salesforce JA3/JA3S Standard	Salesforce Engineering Research <i>Althouse, J., et al. (GitHub open-source standard)</i>	https://github.com/salesforce/ja3
Zeek Network Security Monitor	Lawrence Berkeley National Laboratory <i>Paxson, V. (Computer Networks, 1999)</i>	https://zeek.org/
Scapy Packet Engine	Philippe Biondi & Community <i>Packet creation, dissection & PCAP manipulation</i>	https://scapy.net/
FastAPI Framework	Sebastián Ramírez & Contributors <i>Asynchronous ASGI Web & WebSocket Gateway</i>	https://fastapi.tiangolo.com/
Scikit-Learn Machine Learning	Pedregosa et al. (JMLR 2011) <i>Isolation Forest & Preprocessing Algorithms</i>	https://scikit-learn.org/stable/
NetworkX Graph Analytics	Hagberg, Schult, & Swart (SciPy 2008) <i>Network Structure & Bipartite Graph Analysis</i>	https://networkx.org/

6. Relevant Internet Engineering Task Force (IETF) RFCs

RFC 6455: The WebSocket Protocol

Real-Time Telemetry

Defines full-duplex client-server communication channels over TCP. Utilized by `backend/main.py` to stream high-frequency packet telemetry and alert events directly to the React SOC visualizer without HTTP polling overhead.

<https://datatracker.ietf.org/doc/html/rfc6455>

RFC 1035 & RFC 8499: Domain Names - Implementation and Terminology

DNS Architecture

Specifies DNS query formats, label encodings, and maximum lengths (63 octets per label, 255 octets per name). Foundation for extracting DNS subdomains to detect data tunneling and DGA patterns in `engine/features.py`.

<https://datatracker.ietf.org/doc/html/rfc1035>

RFC 8446: The Transport Layer Security (TLS) Protocol Version 1.3

Encrypted Transport

Standardizes the TLS 1.3 cryptographic handshake. Governs how Client Hello messages, cipher suite advertisements, and supported extensions are structured for JA3 fingerprint generation.

<https://datatracker.ietf.org/doc/html/rfc8446>